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2403A51L01

Batch : 51

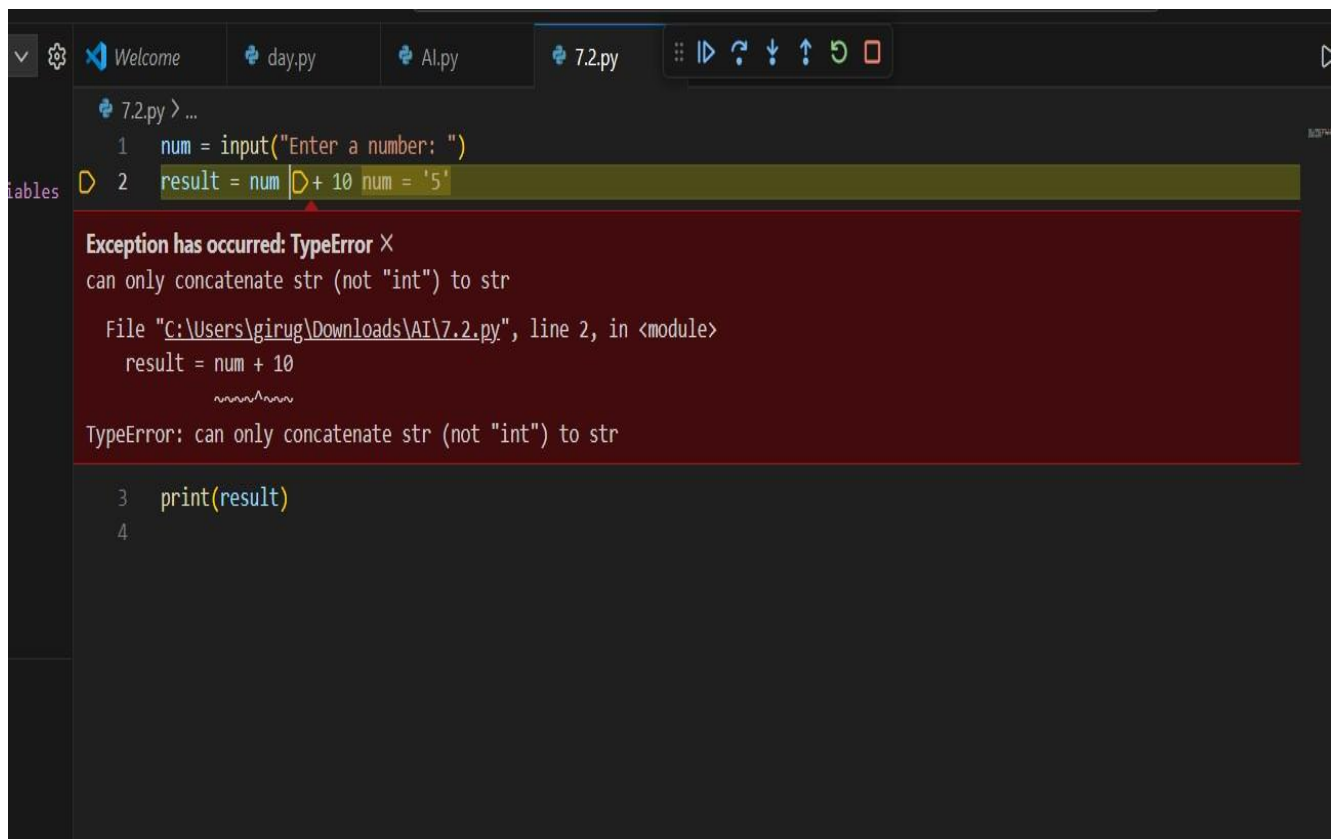
Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs

Task 1 – Runtime Error Due to Invalid Input Type

Prompt:

Write a Python function to determine whether a given number is prime.

Code :

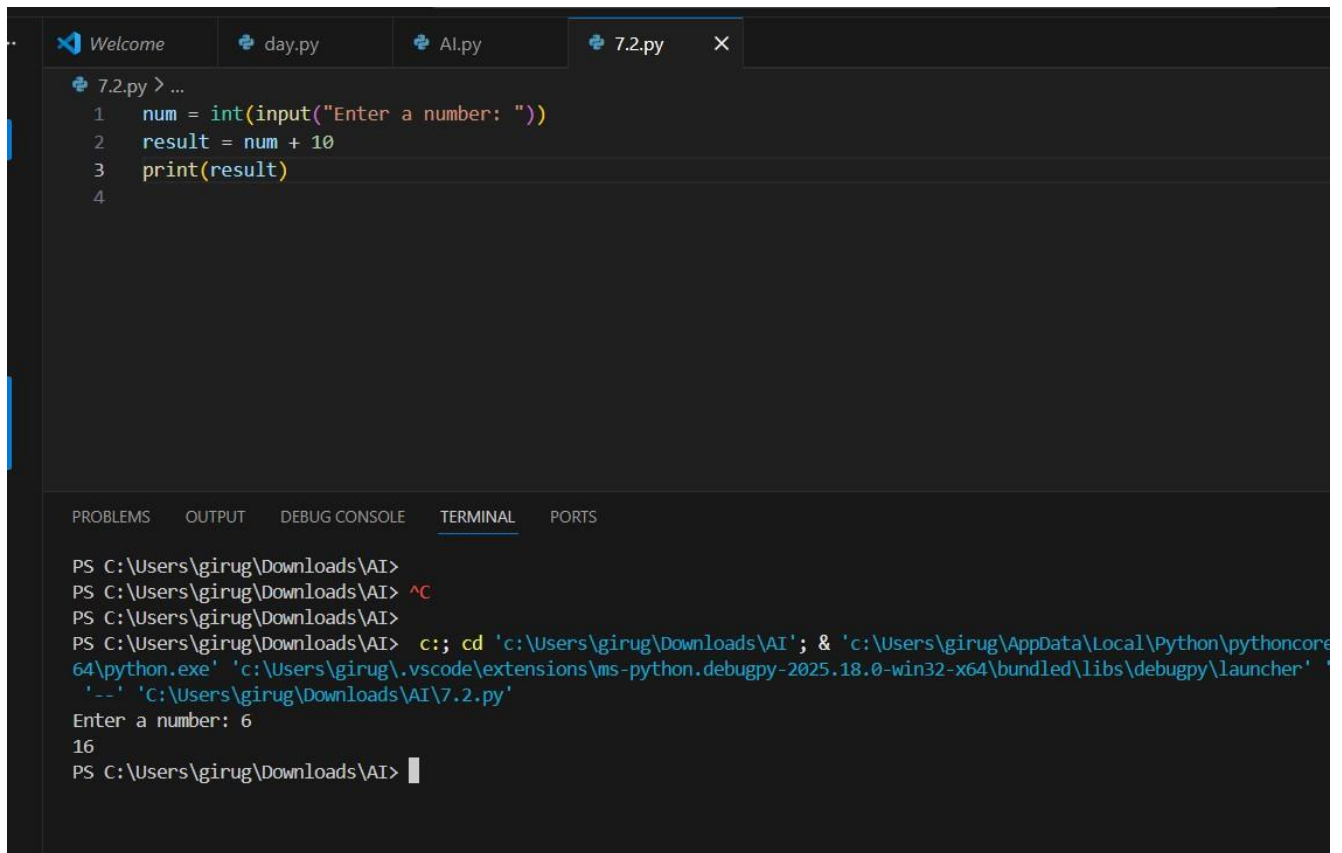


```
7.2.py > ...
1 num = input("Enter a number: ")
2 result = num + 10 num = '5'
3 print(result)
4
```

Exception has occurred: TypeError X
can only concatenate str (not "int") to str

File "C:\Users\girug\Downloads\AI\7.2.py", line 2, in <module>
result = num + 10
~~~~~  
TypeError: can only concatenate str (not "int") to str

## Output:



The screenshot shows a VS Code editor with a file named `7.2.py` open. The code in the editor is:

```
1 num = int(input("Enter a number: "))
2 result = num + 10
3 print(result)
4
```

Below the editor is a terminal window. The terminal shows the command prompt at `C:\Users\girug\Downloads\AI>`. The user enters `^C` to clear the line. Then, they enter `c::; cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '--' 'C:\Users\girug\Downloads\AI\7.2.py'`. The terminal then shows the program's output:

```
Enter a number: 6
16
```

The terminal prompt is now `C:\Users\girug\Downloads\AI>` with a cursor.

**Justification:** The program failed because `input()` returns data as a string, and adding a string to an integer is invalid in Python. The AI corrected this by converting the input to an integer using `int()`, ensuring the arithmetic operation works properly. This type conversion is necessary to match the expected numeric behavior of the program.

## Task 2 – Incorrect Function Return Value

### Prompt:

Generate a function to calculate the sum of elements in a list.

### Code :

```
23
24
25
26
27
28 def square(n):
29     result = n * n
30
31
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\girug\Downloads\AI> c::; cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '51096' '--' 'C:\Users\girug\Downloads\AI\7.2.py'
PS C:\Users\girug\Downloads\AI> 6
6
PS C:\Users\girug\Downloads\AI> 
```

## Output:

```
24
25
26
27 def square(n):
28     result = n * n
29     return result
30 print(square(8))
31
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
> c::; cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '638' '--' 'C:\Users\girug\Downloads\AI\7.2.py'
25
PS C:\Users\girug\Downloads\AI> ^C
PS C:\Users\girug\Downloads\AI>
PS C:\Users\girug\Downloads\AI> c::; cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '639' '--' 'C:\Users\girug\Downloads\AI\7.2.py'
PS C:\Users\girug\Downloads\AI> 8^C
PS C:\Users\girug\Downloads\AI>
PS C:\Users\girug\Downloads\AI> c::; cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '502' '--' 'C:\Users\girug\Downloads\AI\7.2.py'
64
PS C:\Users\girug\Downloads\AI> 
```

Ln 30, Col 15

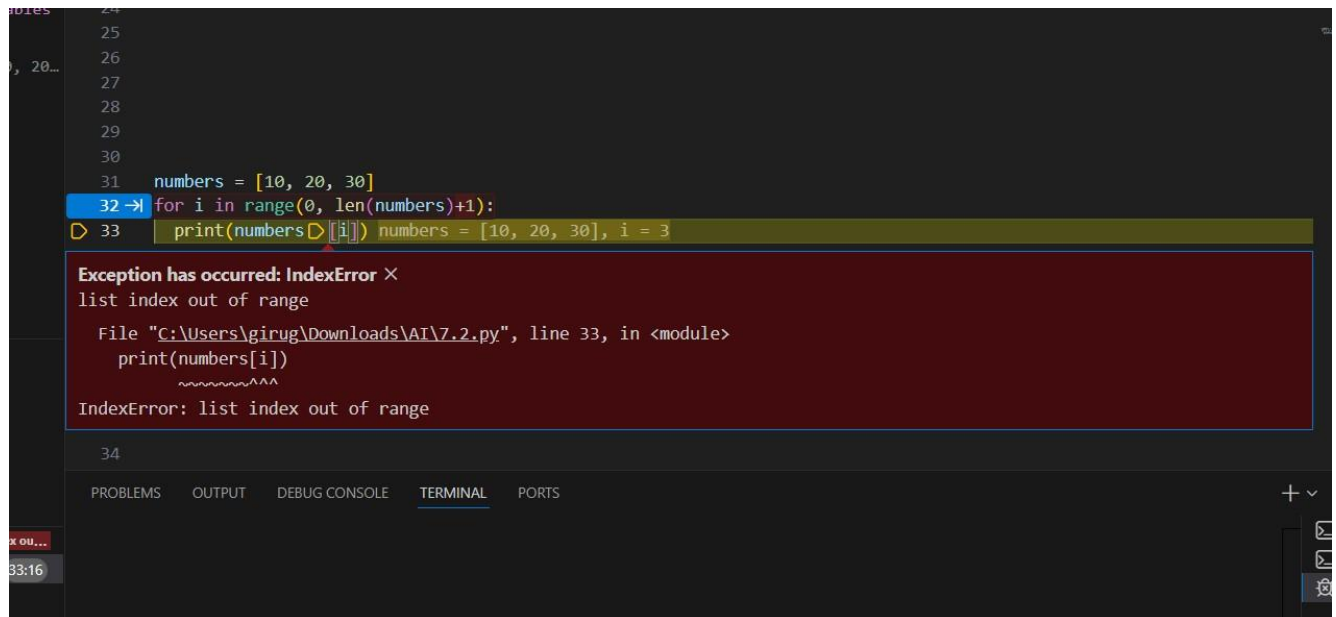
## Justification:

Although the function computed the square internally, it never returned the result, causing the output to be lost. The AI identified the missing return statement and added it, allowing the function to properly send the computed value back to the caller. Returning values is essential for functional correctness and reusability.

### Task 3 – IndexError in List Traversal

Write a Python function that takes an alphanumeric string and returns only the digits.

#### Code :



```
24
25
26
27
28
29
30
31 numbers = [10, 20, 30]
32 → for i in range(0, len(numbers)+1):
33     print(numbers[i]) numbers = [10, 20, 30], i = 3
```

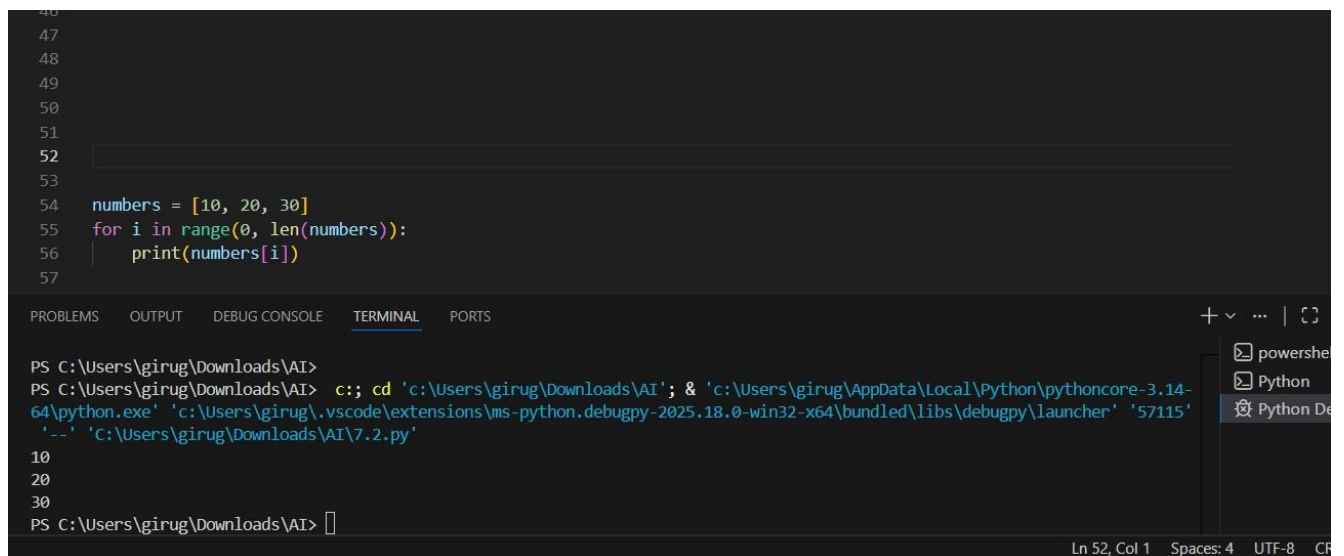
Exception has occurred: IndexError ×  
list index out of range

File "c:\Users\girug\Downloads\AI\7.2.py", line 33, in <module>  
 print(numbers[i])  
 ~~~~~^  
IndexError: list index out of range

34

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Output:



```
40
47
48
49
50
51
52
53
54 numbers = [10, 20, 30]
55 for i in range(0, len(numbers)):
56     print(numbers[i])
57
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\girug\Downloads\AI>
PS C:\Users\girug\Downloads\AI> c:: cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57115' '--' 'c:\Users\girug\Downloads\AI\7.2.py'
10
20
30
PS C:\Users\girug\Downloads\AI>

Ln 52, Col 1 Spaces: 4 UTF-8 CF

Justification:

The loop incorrectly iterated one step beyond the valid index range using `len(numbers) + 1`, causing an `IndexError`. AI fixed the boundary to `range(len(numbers))`, ensuring safe access of all existing list elements. This correction is justified because valid indices only go from 0 to `len(numbers)-1`.

Task 4 – Uninitialized Variable Usage

Prompt :

Write a Python function to count the number of vowels in a given string.

Code :

```
54
55
56
57 if True:
58     pass
59     print(total)
```

Exception has occurred: NameError ×
name 'total' is not defined

File "C:\Users\girug\Downloads\AI\7.2.py", line 59, in <module>
 print(total)
 ^^^^^
NameError: name 'total' is not defined

Output:

```
77
78
79
80 total = 0 # Initializing the variable
81 if True:
82     pass
83
84 print(total)
85
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

'--' 'C:\Users\girug\Downloads\AI\7.2.py'
PS C:\Users\girug\Downloads\AI> ^C
PS C:\Users\girug\Downloads\AI>
PS C:\Users\girug\Downloads\AI> c:: cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '63589'
'--' 'C:\Users\girug\Downloads\AI\7.2.py'
0
PS C:\Users\girug\Downloads\AI> █

Ln 79, Col 1 Spaces: 4 UTF-8

Justification:

The program attempted to print a variable (total) before it had been assigned any value, resulting in a runtime error. AI resolved this by initializing the variable to 0 before use, ensuring the program has a valid reference. Proper initialization prevents undefined behavior and is a fundamental programming requirement.

Task 5 – Logical Error
in Student Grading System Prompt :

write a Python function that takes three numbers and returns the minimum value without using min().

Code :

```
79
80 marks = 85
81 if marks >= 90:
82     grade = "A"
83 elif marks >= 80:
84     grade = "C"
85 else:
86     grade = "B"
87 print(grade)
88
89
90
91
92
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\girug\Downloads\AI> ^C
PS C:\Users\girug\Downloads\AI>
PS C:\Users\girug\Downloads\AI> c;; cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50305' '--' 'C:\Users\girug\Downloads\AI\7.2.py'
PS C:\Users\girug\Downloads\AI> 
```

Output:

```
74
75
76
77
78
79 marks = 85
80
81 if marks >= 90:
82     grade = "A"
83 elif marks >= 80:
84     grade = "B"
85 else:
86     grade = "C"
87
88 print(grade)
89
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C
PS C:\Users\girug\Downloads\AI> ^C
PS C:\Users\girug\Downloads\AI>
PS C:\Users\girug\Downloads\AI> c;; cd 'c:\Users\girug\Downloads\AI'; & 'c:\Users\girug\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\girug\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '64291' '--' 'C:\Users\girug\Downloads\AI\7.2.py'
PS C:\Users\girug\Downloads\AI> 
```

Justification:

The conditions for assigning grades were incorrectly ordered, making the program assign a wrong grade for certain mark ranges. AI fixed this by arranging the conditions in a logically descending order (A → B → C), ensuring accurate evaluation. Correct conditional structure is essential for producing correct program decisions.